

Module 5: Genetics, control and homeostasis

5.1 Genetics in the twenty first century

The aim of this section is to develop an understanding of the fundamental principles of heredity, which are illustrated by reference to model organisms and to the inheritance of genetic diseases.

Learners gain an understanding of the regulation of the genome by a number of factors, and the concept of epigenetics will also be considered in relation to the increase in diseases such as Type 2 diabetes.

Some of the practical techniques used to analyse and manipulate DNA are also considered, as well as the applications of these techniques. The potential uses of DNA and RNA technology in the treatment of disease are considered, as is the profound ethical

implications of gene technologies and the role of genetic counsellors.

The use of gene technologies to catalogue the accumulation of different genetic information in populations and the implications for speciation can also be considered.

This section provides opportunities for the use of statistical tests in analysing data.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.